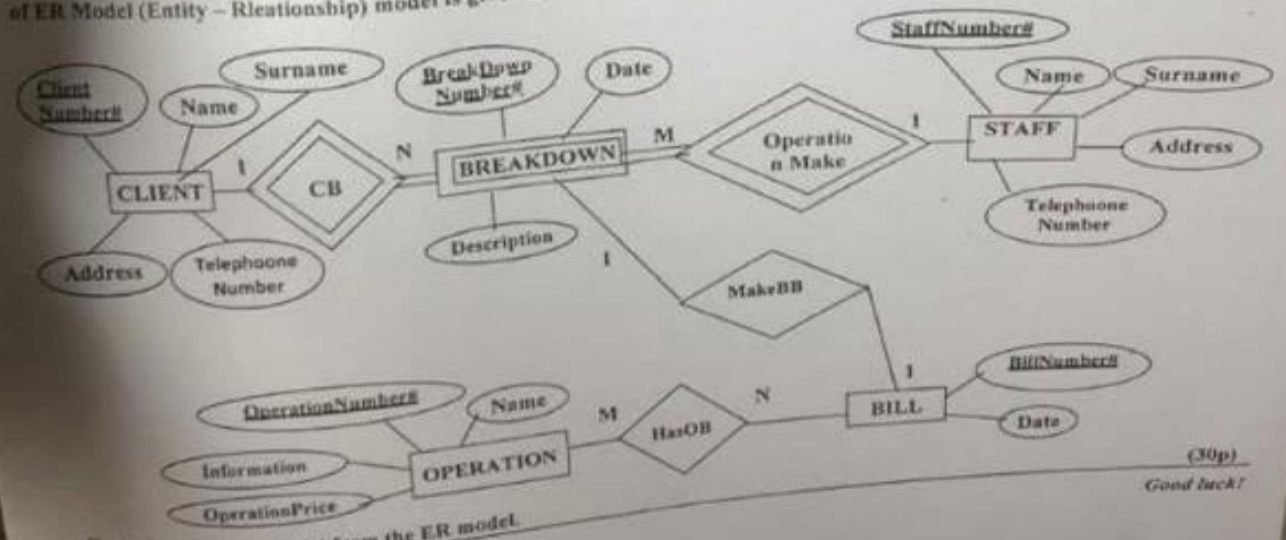


- a) Name the type(s) of attributes (12p)
 b) Name the type(s) of relationships (09 p)
 c) Define "Specialization" and "Generalization" in EER diagrams (06 p)

2) Doctors of the World (DW) is an international organization dedicated to implement health care plans in poor countries. DW needs to design a database to record information to fulfill the following specifications:

- Doctors are identified by their id. It is required for each doctor to record her/his name, birthday, country of citizenship and sex.
- A doctor has always at least one medical specialty. A medical specialty is identified by a code and has a name. The doctors are ranked in each of her/his specialties according to their years of experience in that specialty.
- The organization collaborates with international agencies to implement and deploy health plans in some countries. Each international agency is identified by an id and has a name. Countries are identified by their names. For each country, its population and its poverty index must be also stored.
- Each health plan is implemented in a given country, and the name of the health plan is uniquely associated among the health plans of that country. Each health plan has a budget and is carried out during a period of time. A health plan is always sponsored by one or more international agencies. Each international agency provides part of the budget of the health plan. The total budget provided by the agencies is never less than the health plan budget.
- A country attended by the organization needs some medical specialties. A health plan designed for a given country covers some of its needs for medical specialties.
- A doctor may participate in several health plans. The period of time that a doctor participates in a health plan must be recorded. A doctor cannot participate in two health plans simultaneously. A doctor never participates in a health plan more than once. The participation of a doctor in a health plan never exceeds the duration of the plan.
- Each health plan includes at least 7 doctors; one of who is the coordinator of the health plan. The coordinator of the plan always works for during the duration of the plan. A doctor involved in a health plan must have one of the specialties required by the health plan.

3) A combi service, information of repair and maintenance of the combi service is wanted to keep in a database and its design of ER Model (Entity - Relationship) model is given below.



Create the Relation Model from the ER model.

(30p)
 Good luck!

YU – Faculty of Chemical and Metallurgical Engineering, Questions and Answers Sheet

Student Name and Surname: [REDACTED]

Students Number: [REDACTED]

Section: MATHEMATICAL ENGINEERING DEPARTMENT

Course Name: DATABASE MAN. and FILE STR.1

Course Instructor Name and Surname: Ayta ŞAYLI and Armagan ELİBOL

Group Number: [REDACTED]

Exam Duration: [REDACTED]

min: [REDACTED]

Exam Date: 06/06/2018

Examination: [REDACTED]

Signature: [REDACTED]

NOTE CHART

1. q	2. q	3. q	Total

1) Please use the following table named as COMPLAIN where RegNo# is the primary key.

RegNo#	Date	CustNo#	Name	Comp No#	Type#	TypeName	Id#	CName	Dept No#	Dept	Bld#	Behaviour	How Many	Total Comp	CustTelNo	Living Area
1	01/01/2012	10000	Derya Güç	1	C1	Secretary	1000	Kadriye Aydın	1	Product	O	Oral	3	1	2123333333	Beşiktaş
				2	C2	Assistant	500	Gunduz Selder	2	Sales	W	Written	1			
				3	C3	Supervisor	100	Cem Derya	1	Product	F	Fight	2			
				4	C4	Administrator	1500	Yazul Sezer	1	Product	F	Fight	1		2161111111	Kadıköy
2	01/03/2012	15001	Ceylan Er	1	C2	Assistant	500	Gunduz Selder	2	Sales	W	Written	2	3		
				2	C3	Supervisor	100	Cem Derya	1	Product	F	Fight	1			
3	15/05/2013	10000	Derya Güç	1	C1	Secretary	1000	Kadriye Aydın	1	Product	W	Written	1	1	2123333333	Beşiktaş

- Which one is the first normal form of COMPLAIN
- A) COMPLAIN (RegNo#, Date, CustNo#, Name, TotalComp, CustTelNo, LivingArea)
- B) COMPLAIN (RegNo#, Date, CustNo#, Name, CompNo#, Type#, TypeName, Id#, CName, DeptNo#, Dept, Bld#, Behaviour, HowMany, TotalComp, CustTelNo, LivingArea)
- C) COMPLAIN (RegNo#, Date, CustNo#, Name, CompNo#)
- D) None

2. The following ENTRY table is given, which normalisation form should be applied first?

EntryNo#	Name	Sport#	SName	ClassNo	Teacher#	TName	TAddress	TTelNo	PricePerHour
100	Gül Er	1	Tennis	4	223	A. Ak	Beşiktaş	333 33 33	50
101	Cem Öz	2	Squash	7	225	D. Deniz	Kadıköy	444 44 44	75
102	Gül Er	1	Tennis	5	234	A. Şahin	Taksim	555 55 55	50

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF
3. After applying the normalisation in the previous, what is the result?
- A) ENTRY(EntryNo#, Name, Sport#, ClassNo, Teacher#, PricePerHour), SPORT(Sport#, SName), TEACHER(Teacher#, TName, Address, TtelNo)
- B) ENTRY(EntryNo#, Name, ClassNo, PricePerHour), SPORT(EntryNo#, Sport#, SName), TEACHER(EntryNo#, Teacher#, TName, Address, TtelNo)
- C) ENTRY(EntryNo#, Name, ClassNo, PricePerHour), SPORT(Sport#, SName, EntryNo#), TEACHER(Teacher#, TName, Address, TtelNo, EntryNo#)
- D) ENTRY(EntryNo#, Name, ClassNo, PricePerHour), SPORT(Sport#, SName), TEACHER(Teacher#, TName, Address, TtelNo, EntryNo#)

1	SQUASH	30	5
2	TENNIS	10	1
2	SQUASH	15	3
3	SWIMMING	10	5
3	TENNIS	20	

A) 1NF

B) 2NF

C) 3NF

D) BCNF

5. After applying the normalisation in the previous question, what is the result?

- A) SCHOOL_DETAIL(No#, SportType#, COST(SportType#, NumberofPeople, PricePerHour))
 B) SCHOOL_DETAIL(No#, NumberofPeople), COST(SportType#, PricePerHour, No#)
 C) SCHOOL_DETAIL(No#, SportType#, NumberofPeople), COST(SportType#, PricePerHour)
 D) SCHOOL_DETAIL(No#, SportType#), COST(SportType#, PricePerHour), PERSON(SportType# NumberofPeople)

6. When are the normalisation forms not needed for?

- A) Eliminate redundancy
 B) Remove duplicates

C) Avoid anomalies

D) Force integrity constraints

7. Which of the options below should not be included in the first normal form?

- A) Multi-valued attributes

B) Repeating Group - Nested relation

C) Composite Attributes

D) All

8. Which one is the one of the update anomalies?

- A) Insertion

B) Deletion

C) Modification

D) Wastes storage

Given two relations R1 and R2 (union-compatible), where R1 contains N1 tuples and R2 contains N2 tuples, and $N2 > N1 > 0$, the maximum and minimum possible sizes (in tuples) for the resulting relation produced by each of the following relational algebra expressions.

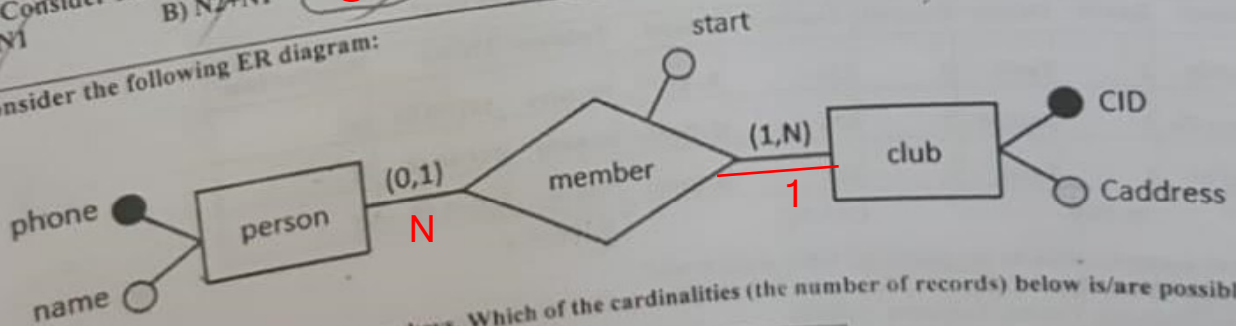
9) Consider $R1 \cup R2$, What is the maximum number of records ('tuples')?
 A) $N1 * N2$
 B) $N1 + N2$
 C) $N1$
 D) $N2$

10) Consider $R1 \cap R2$, What is the minimum number of records ('tuples')?
 A) $N1$
 B) $N2$
 C) 0
 D) $N2 - N1$

11) Consider $R1 - R2$, What is the maximum number of records ('tuples')?
 A) $N1$
 B) $N2 - N1$
 C) 0
 D) $N2$

12) Consider $R1 \cap R2$, What is the minimum number of records ('tuples')?
 A) $N1$
 B) $N2 + N1$
 C) 0
 D) $N2$

Consider the following ER diagram:



Where phone and CID are primary keys. Which of the cardinalities (the number of records) below is/are possible? When your answer is not possible, explain briefly why.

Question No	person	member	club	Is it possible?
13.	5	0	8	A. Yes B. No
14.	5	7	8	A. Yes B. No
15.	5	0	5	A. Yes B. No
16.	5	10	5	A. Yes B. No
17.	11	3	4	A. Yes B. No
18.	11	9	4	A. Yes B. No

Her kulüp var olmak için en az 1 üyeye ihtiyaç duyar.
 Kişi kulübe üye olmak zorunda değildir.
 Kişi en fazla 1 kulübe üye olabilir.
 Kulüpte 1den fazla üye olabilir.

Use the following relational model of GAME database, answer these questions in this section.

USER (UserId#, FirstName, InitialName, FamilyName, BirthDate)
 CLUB (ClubCode#, CName, CAddress, CTelNo, CSet-Up-Date, CSet-Up-UserId#, Manager_UserId#, Country)
 TEAM (TeamNo#, TeamName, TSet-Up-Date, TSet-Up-UserId#, Captain_UserId#, ClubCode#, WorkArea)
 TEAM_MEMBER (TeamNo#, UserId#, JoinDate)
 STADIUM (Stadium#, SName, SAddress, Day, Month, Year)
 MATCH (Stadium#, Date#, Game#, A_TeamNo#, ScoreA, B_TeamNo#, ScoreB)

Each club must have one set-up user who cannot be related to another club, this is similar for club managers. Every team must have a captain who works only one team at the same time. Clubs have different teams with distinct names. A game cannot be played because of the forecast or not the attendance of less than 10 players per team and 3 referees.

32. Select the total number of relationships.
 A) 11 B) 10 C) 9 D) 8
33. Select the total number of relationship(s) between USER and CLUB.
 A) 3 B) 2 C) 1 D) 0
34. Select the relationship type of the relationship(s) between USER and CLUB.
 A) 1:N B) M:N C) 1:1 D) N:0
35. Select the total number of relationship between USER and TEAM.
 A) 3 B) 2 C) 1 D) 0
36. Select the relationship type of the relationship between USER and TEAM.
 A) 1:N B) M:N C) 1:1 D) N:0
37. Select the total number of relationship(s) between MATCH and TEAM.
 A) 0 B) 1 C) 2 D) 3
38. Select the relationship type of the relationship between MATCH and TEAM.
 A) 1:N B) M:N C) 1:1 D) N:2
39. Select the total number of relationship(s) between TEAM and TEAM_MEMBER.
 A) 4 B) 3 C) 2 D) 1
40. Select the relationship type of the relationship between TEAM and TEAM_MEMBER.
 A) 1:N B) M:N C) 1:1 D) N:0
41. Select the relationship type of the relationship between CLUB and TEAM.
 A) 1:N B) M:N C) 1:1 D) N:0
42. Select the total number of weak relationships in ER Model of MATCH.
 A) 5 B) 4 C) 3 D) 1
43. Select the weak entities in ER Model of MATCH.
 A) CLUB - TEAM B) TEAM_MEMBER - MATCH C) USER - STADIUM D) TEAM-MATCH-CLUB

Answer each statement if it is true, choose A; otherwise choose B for false.

Question No	Statement	True (T) / False (F)
44.	Given 2 union-compatible tables M and N with 100 rows and 20 rows respectively. The minimum number of rows in $(M - (M - N))$ is 20.	True
45.	The primary key of a table is made up columns x and y. In a given row, either x or y can be null, but not both.	False
46.	The columns of a relation are sometimes called "tuples."	False
47.	A relation can have only one candidate key.	False
48.	A weak entity is an entity that cannot exist in the database without (and is logically dependent upon) another type of entity also existing in the database.	True
49.	In the relational model, cardinality is referred to as the number of attributes in a relation.	False
50.	In case of entity integrity, primary key can be null.	False

tuples: Satır, row
 Domain: Kolon, column